PRODUCTION SPECIFICATION

Laju

// FLEET MANAGEMENT & LIVE DISPATCH ENGINE

Laju is a mission-critical fleet telemetry and predictive dispatch platform created for Indonesia's largest inter-island logistics operators, serving 2,400 active long-haul trucks.

PRIMARY ROLE: LEAD PRODUCT & INTERACTION DESIGNER



CLIENT / ORG	CHRONOLOGY	DISCIPLINE	DELIVERABLES
Laju Logistik Nusantara	2024	UIUX	Multi-modal Driver Telematics App · Fleet Dispatch Cockpit (Web & Dual-screen)

EXECUTIVE OVERVIEW

The Strategic Problem Space

Laju is a mission-critical fleet telemetry and predictive dispatch platform created for Indonesia's largest inter-island logistics operators, serving 2,400 active long-haul trucks.

OPERATIONAL CHALLENGE

Dispatchers were overwhelmed by 14 separate alert monitors, causing average emergency response lag times of 8.2 minutes and driver communication breakdown in low-connectivity routes.

ENGINEERED INTERVENTION

Formulated the "Dark Cockpit" design paradigm: reduced visual cognitive load by 60%, introducing context-aware priority surfacing, high-contrast night modes, and tactile one-thumb mobile workflows.

METHODOLOGY ROADMAP

Phased Execution Model

STEP 01 // FIELD RIDE-ALONGS & DISPATCH ETHNOGRAPHY

Conducted 48 hours of cab ride-alongs along the Trans-Java tollway and shadowed 12 dispatcher shift handovers under live incident conditions.

STEP 02 // COGNITIVE URGENCY TIERING & FLOW MAPPING

Categorized 140 telemetry signals into Ambient, Actionable, and Immediate Critical tiers to eliminate alert fatigue.

STEP 03 // TACTILE MOBILE & DARK-COCKPIT WIREFRAMING

Prototyped high-contrast UI screens optimized for single-hand use by drivers wearing heavy work gloves at high speeds.

STEP 04 // TOKENIZED DESIGN SYSTEM & USABILITY VALIDATION

Built the "Vektor" token library in Figma and code, validated through dual-screen simulator tests with 24 active dispatchers.

// ARCHITECTURAL CREED

"In high-stress logistics, good UI isn't decorative — it is a safety barrier that prevents driver fatigue and catastrophic routing failures."

DESIGN SYSTEMS & TOKEN SPECIFICATION

Atomic Tokens & Cognitive Flow

SYSTEM COLOR PALETTE TOKENS

Cockpit Slate
#050811

Telemetry Blue
#0057FF

Warning Amber
#F59E0B

Critical Crimson
#EF4444

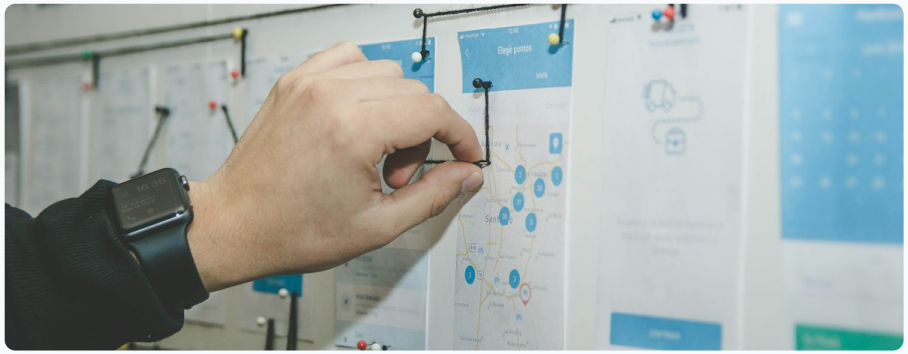
TYPOGRAPHY SPECIMEN TABLE

Data Value / Telemetry
JetBrains Mono · 20px / Bold

Cockpit Navigation
Plus Jakarta Sans · 14px / SemiBold

Incident Narrative
Plus Jakarta Sans · 16px / Regular

USER JOURNEY & WIREFRAME FLOW



FINAL INTERACTION ARTIFACTS

High-Fidelity Telematics Interfaces

PRODUCTION SCREEN 01



PRODUCTION SCREEN 02



MEASURED PRODUCTION TELEMETRY

Quantified Business & Architectural Results

-65%	98.2%	1.4%	89.5 / 100
DISPATCH REACTION TIME	DRIVER TASK COMPLETION	DISPATCHER ERROR RATE	SYSTEM SUS SCORE



LIVE INTERACTIVE DEPLOYMENT

Scan QR code to inspect the live web case study, interactive components, and telemetry streams.

URL: http://localhost:3000/projects/laju-fleet-dispatch
DEMO: https://laju.design.preview

AUTHOR & ARCHITECT

Mahesa Alghifari

UI/UX Designer · Fullstack Engineer · Graphic Designer

mahesa@alghifari.dev
West Java, Indonesia · © 2026